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Armoury element for the protection of a structural material and/or load-carrying element

Abstract

An armoury assembly is provided for the protection of a structural material and/or load-carrying element having a longitudinal axis. The armoury assembly is provided longitudinally surrounding the structural material and/or load-carrying element to be protected. The armoury assembly comprises at least two different layers, one being an energy-absorption matrix, the other layer being made of a metal, an alloy or a fibre reinforced polymer having a thickness less than the energy-absorption matrix. Two or more longitudinal channels are provided to the armoury assembly, wherein the channels are substantially parallel to the longitudinal axis of the structural material and/or the load-carrying element.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) The present invention relates to the technical field of cables, in particular to stay cables, but it is also equally applicable to other technical fields relating to architectures including constructions and buildings. Constructions such as masts, towers, bridges, footbridges and roofs for stadium, where their essential and functional components (columns, beams or rods and the like) are to be protected from external and sudden threats, for instance from fire outbreak, targeted cutting by grinder or torch, sudden explosion or targeted blast.

BACKGROUND OF THE INVENTION

(2) In recent time, an increasing number of fire outbreaks and terrorist attacks have shown that the effects of fire and blast loads on constructions and buildings are serious matters that should be taken into consideration, whether in the initial design process, during the construction process or after the completion of the construction.

(3) Although these kinds of attacks are man-made disasters and are usually exceptional cases, its potential loss from fire or blast (blast load) are in fact needs to be carefully calculated just like other risks such as earthquake and wind loads.

(4) For these reasons, damage to the assets, loss of life and social panic are factors that have to be minimized if those threats cannot be stopped.

(5) Patent document GB 686804A relates to a protection armour for electric cables. It discloses that the electrical cable comprises an external protective armour constituted of metallic braid. The component elements are entirely and individually coated with a tough, flexible and dielectric material, wherein the material is a plastic capable of resisting corrosion, abrasion and not inflammable.

(6) Another patent document U.S. Pat. No. 2,909,336 relates to an armoured cable, for instance an armoured subaquatic cable, in which the armour is formed by a plurality of wires wound helically around a core of the cable. The armoured cable comprises a bunch of metal filaments formed of copper, aluminium or their alloys, being wrapped or encased in layers of fabric, rubber, impregnated paper, bitumen impregnated jute and sheath to form a protective shield over said cable.

(7) Although these cables are being provided with an armour protection, they are not ideal for the protection for the purpose of the present case where the elements to be protected should be safe from fire outbreaks and/or blast. Although designing the structures to be fully fire- and blast resistant is not a realistic and an economical option, the need for such an armoury element for precautionary purpose remains high.

SUMMARY OF THE INVENTION

(8) The inventors of the present invention have found out effective remedies for the above-discussed problems with the current engineering and architecture knowledge such that the new and existing constructions and buildings can be equipped with the protective assemblies and elements according to the present invention to mitigate the effects of external threats including fire outbreaks and sudden blast.

(9) In a first aspect, present invention relates to an armoury assembly for the protection of a structural material and/or load-carrying element having a longitudinal axis, wherein the armoury assembly is provided longitudinally surrounding the structural material and/or load-carrying element to be protected, wherein the armoury assembly comprises at least two different layers, one being an energy-absorption matrix, preferably confined or supported within and by the other, being made of a metal, an alloy or a fibre reinforced polymer having a thickness less than the energy-absorption matrix, wherein two or more longitudinal channels are being provided to the armoury assembly, wherein the channels are substantially parallel to the longitudinal axis of the structural material and/or the load-carrying element.

(10) In a second aspect, present invention relates to a stay cable pre-fitted or retro-fitted with an

armoury assembly of the present invention.

(11) In a third aspect, present invention relates to a structural material of a construction or a building, wherein its component such as column, rod or beam is pre-fitted or retro-fitted with an armoury assembly of the present invention.

(12) In one embodiment of the present invention, the armoury assembly comprises two or more channels, wherein at least one of the channel has a geometry which permits threading of a single wire or strand element thereto. This has the advantageous of exerting compressing forces (e.g. longitudinally, radially and etc.) to the armoury assembly **100**.

(13) In yet another embodiment of the present invention, the energy-absorption matrix comprises a solid filler such as concrete, ashcrete, polymer-concrete or timbercrete having a compressive strength of at least about 20 MPa and/or at most about 300 MPa, preferably at most about 120 MPa. Concrete has the advantage of easy availability for large-scale production. Ashcrete is a concrete alternative that uses fly ash instead of traditional cement. By using fly ash, a by-product of burning coal, 97 percent of traditional components in concrete can be replaced with recycled material, hence it is more environmentally. Polymer-concrete is concrete matrix reinforced by polymeric fibres which present higher ductility and fire resistance, permitting higher energy absorption and better protective capabilities. Timbercrete is a building material made of sawdust and concrete mixed together. Since it is lighter than concrete, it reduces transportation emissions, and the sawdust both reuses a waste product and replaces some of the energy-intensive components of traditional concrete. Due to its light-weight, Timbercrete could be an option for the armoury assembly for use in stay cable for instance.

(14) In a further embodiment, at least some or most of the channels are being provided to the energy absorption matrix to accommodate one or more wire or strand elements thereto, wherein the wire or strand element can be arranged in such a way to exert compressing force radially along the longitudinal axis. This allows the armoury assembly to be strengthened by the synergistic effect from the energy absorption matrix and the wire/strand elements.

(15) In one preferred embodiment, pipe element being provided to the longitudinal channel for receiving wire or strand element accommodated thereto, wherein the wire or strand element extends axially or a helical along the longitudinal axis, for instance in a single-, double- or multiple-helical manner e.g. laying in both left handed and right handed direction.

(16) In another preferred embodiment, the layer made of metal, alloy or fibre reinforced polymer comprises a plurality of patch-like elements that are being assembled, connected and tightened to each other e.g. by use of strand or wires such as to permit later retrofit of critical member by such protection, preferably arranged in such a way to exert a compression force towards the central axis of the structural material and/or load-carrying element.

(17) In yet another embodiment, the armoury assembly comprises an outer layer and an inner layer, wherein the layers being made of a metal, an alloy, or fibre reinforced polymer. High temperature resistance metal or alloy can be used to for such layers. Alternatively, fibre reinforced polymers can be selected due to its light weight property. In a further embodiment, the inner layer can be made of fibre reinforced polymer and the outer layer can be made of metal or alloy.

(18) According to one specific embodiment, the pipe element comprises the inner layer or can be considered to be identical as the inner layer. In this embodiment, the inner layer is in form of a pipe such that it is capable of receiving structural material and/or load-carrying element to be protected.

(19) According to another particular embodiment, the inner layer is provided to surround longitudinally at least some of the load-carrying elements such as strand bundles of tensile elements, wherein the each inner layer surrounding longitudinally the load-carrying elements to be protected preferably has the same thickness as the outer layer. This embodiment has the advantage that some of the load-carrying elements can be served as a sacrificial component (if no inner wall or layer surrounding them) while the overall structure integrity of the elements to be protected remains intact.

- (20) In one particularly preferred embodiment, the energy absorption matrix is sandwiched between the outer layer and the inner layer. This configuration gives an optimum protection for the structural material and/or load-carrying element to be protected.
- (21) In yet a preferred embodiment, a plurality of the longitudinal channels having approximately about the same diameter are provided to the armoury assembly for accommodating wire or strand element and/or load-carrying element.
- (22) In a further embodiment, the channels being provided to the armoury assembly are arranged randomly or distance approximately equally from each other. Such arrangement allows an optimal protection from fire and blast threats. For instance, the distance between each longitudinal channel is preferably between 0 cm and 50 cm, preferably between 0.2 cm and 25.0 cm, or preferably between 0.2 cm and 2.0 cm.
- (23) In one specific embodiment, the outer layer being made of a material having a yield strength of at most about 2000 MPa and/or at least about 200 MPa, and the inner layer is made of a material having a yield strength of at most 2000 MPa and/or at least about 200 MPa.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1a is a perspective view of the armoury assembly according to a first embodiment of the present invention.
- (2) FIG. 1b is a longitudinally half-sectioned perspective view of the armoury assembly according to a first embodiment of the present invention.
- (3) FIG. 1c is a plan view of the first embodiment of the present invention.
- (4) FIG. 2a is a cross sectional view of the armoury assembly according to a second embodiment of the present invention.
- (5) FIG. 2b is a perspective view of a second embodiment according to the present invention demonstrating the retro-fitted principle of how the armoury assembly is used to protect the load-carrying elements of a stay cable.
- (6) FIG. 2c is a perspective view of a second embodiment according to the present invention demonstrating the retro-fitted principle of how the armoury assembly is used to protect a structural material.
- (7) FIG. 3a is a perspective view of the armoury assembly according to a third embodiment of the present invention.
- (8) FIG. 3b is a longitudinally half-sectioned perspective view of the armoury assembly according to a third embodiment of the present invention.
- (9) FIG. 3c is a plan view of the armoury assembly according to a third embodiment of the present invention.
- (10) FIG. 4 is a perspective view of a third embodiment according to the present invention demonstrating a pre-fitted principle of how the armoury assembly is used to protect load-carrying elements.

DETAIL DESCRIPTION OF THE INVENTION

- (11) Several preferred embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, a detailed description of known functions and configurations incorporated herein has been omitted for conciseness. To this end, it is pointed out that different features from different embodiments can be selected to be combined together within capability of a skilled person in the art.
- (12) FIG. 1a shows an armoury assembly **100** according to a first embodiment of the present invention. In this preferred embodiment, the armoury assembly **100** comprises at least two layers, wherein a layer **10** completely encircles an energy absorption matrix layer **20**. This layer **10** defines

the contour of the armoury assembly **100**, and is suitably made of a metal, an alloy or a fibre reinforced material.

(13) The energy absorption matrix layer **20** has a thickness larger than the outer layer **10**. Said energy absorption matrix **20** comprises a solid filler, for instance made of a concrete or the like, such as ashcrete (from fly ash instead of cement) or polymer-concrete or timbercrete. These kind of materials are suitable for absorbing shock waves energy resulting from sudden blast and the matrix is also resistant to high temperature caused by for instance fire. It is also foreseen that the energy absorption matrix **20** can be provided in two, three or more layers. Such multiple layers of energy absorption matrix **20** could increase the blast resistance of various types of direct impacts and shock waves.

(14) FIG. **1b** illustrates a longitudinally half-sectioned perspective view of the armoury assembly **100**, which has a predominantly cylindrical shape. To this end, it can be foreseen that any other shape (e.g. square, rectangular, oval or irregular shapes) can also be protected by the armoury assembly **100** of the present invention, with little or no modification required.

(15) As clearly shown in this FIG. **1b**, the armoury assembly **100** comprises a layer **10** which is at the outermost of the armoury assembly **100**, an energy absorption matrix **20** and a plurality of channels **30**, namely a channel **30a** having a larger diameter in the central longitudinal axis of the armoury assembly **100** and two channels **30b** having a smaller diameter (on the far left side). The channel **30a** in the central position is suitable for accommodating elements to be protected. The two channels **30b** having a smaller diameter compared to the channel **30a** in the central position are provided to accommodate wire or strand elements **75**. These elements **75** can exert a compressing force radially to the armoury assembly **100**. These channels **30b** are provided helically for instance to the energy absorption matrix **20**, as can be seen in the half section of the armoury assembly **100** where four partially cut-through channels **30b** are shown.

(16) In this embodiment, the armoury assembly **100** can be retro-fitted to protect the structural material and/or load-carrying elements which have been completely installed or constructed from external threats. In order to achieve this purpose, the armoury assembly **100** has a “casing-like” structure where the elements to be protected can easily be encased and shielded by the armoury assembly **100** from external threats as described. In other words, the central part of the armoury assembly forms a channel **30** having a large diameter for housing the structural material (e.g. column) and/or load-carrying element (e.g. tensile members of a stay cable). Such configuration allows the elements to be protected do not require any post-constructional modification (or only little structural modifications) for the installation of the armoury assembly **100**. Of course, it can also be foreseen that such armoury assembly **100** can also be pre-fitted to the structural material and/or load-carrying element to be protected before the installation or construction.

(17) FIG. **1c** is a plan view of the first embodiment. This embodiment of the armoury assembly **100** comprises an inner diameter **N** and an outer diameter **M**. The inner diameter **N** of the armoury assembly **100** may range from 50 mm to 400 mm, typically 100 mm to 350, preferably 150 mm to 250 or more preferably around 200 mm. The outer diameter **M** of the armoury assembly **100** of the present invention may range from about 100 mm to 800 mm, typically from about 200 mm to 500 mm, preferably from about 250 mm to 400 mm or preferably from about 320 mm to 350 mm. In one most preferred embodiment, the inner diameter **N** and the outer diameter **M** of the armoury assembly **100** are about 200 mm and 350 mm, respectively. The structural material and/or the load-carrying element (e.g. housed in a pipe) to be protected may have a diameter ranging from about 40 mm to 380 mm, typically from about 100 mm to 280 mm, preferably from about 130 mm to 230 mm or more preferably from about 170 mm to 200 mm.

(18) FIG. **1c** also illustrates that apart from the channel **30a** located in the central position of the armoury assembly **100**, a plurality of channels **30b** are additionally provided to the energy absorption matrix **20**, wherein the diameter of these channels **30b** are generally much smaller than the diameter of the channel **30a** located in the central position. These channels **30b** typically have a

small diameter, for instance ranging from about 5 mm to 80 mm, preferably from about 10 mm to 50 mm, preferably from about 15 mm to 30 mm or in most cases about 25 mm. These channels **30b** are provided to receive wires or strand elements **75** such that compressing or tensioning force can be exerted radially to the armoury assembly **100**. This can be achieved by tightening the wire or strand elements **75** longitudinally around the elements to be protected. Moreover, it is disclosed herewith that these channels **30b** are distributed in the entire circumferential of the armoury assembly **100**, as can be seen in the plan view of the FIG. **1c**. The distribution of the channels **30b** can either be random or provided equally spaced from each other.

(19) To this end, it is mentioned that these parameters of the inner diameter **N**, the outer diameter **M** of the armoury assembly **100** as well as the diameter of the channel **30b** for receiving wire or strand elements **75** are applicable to all embodiments of the present invention.

(20) FIG. **2a** shows another variant of the embodiment of the present invention, wherein in addition to the outer layer **10** at the outermost surface of the armoury assembly **100**, an inner layer **40** can further be provided to the armoury assembly **100**, wherein the energy absorption matrix **20** is sandwiched or confined by these two layers, namely the outer layer **10** and the inner layer **40**. The channels **30a**, **30b** in this second embodiment are otherwise similar as described in the first embodiment.

(21) The channel **30a** of the armoury assembly **100** according to the first and second embodiments can be used to accommodate load-carrying elements **85**, for example of a stay cable **95**, as shown in FIG. **2b**, or can be used to accommodate structural material **115** of a construction or a building such as column, as illustrated in FIG. **2c**. In both FIGS. **2b** and **2c**, the channels **30b** having a smaller diameter are being provided to the energy absorption matrix **20** accommodate wire and strand elements **75**.

(22) To this end, it is mentioned that the load-carrying elements **85** (e.g. tensile members) are typically housed within a pipe of a stay cable **95**. Moreover, the armoury assembly **100** of all embodiments of the present invention can be customised such that its inner and outer diameters can be retro-fitted to accommodate different elements to be protected. The armoury assembly **100** of the present invention can be provided for instance in two half sections, and later be connected, tightened and/or sealed to form the armoury assembly **100** as claimed presently. Alternative, the armoury assembly **100** can also be provided in three, four, five or more pieces, assembled, tightened and/or sealed together forming the armoury assembly **100** as described in the first and second embodiments.

(23) The armoury assembly **100** forming from two half, three or more sections allows an easy mounting to the elements to be protected. Nevertheless, such characteristic weakens the capability of the armoury assembly **100** from shielding of different threats such as fire, blasts, mechanical cutting, thermal torch cutting and etc., as gaps or connecting points of the armoury assembly **100** due to the sections are more susceptible to the above-mentioned threats. Therefore, it is foreseeable and preferred that the armoury assembly **100** is provided as one piece e.g. one rounded piece (without connecting sections/pieces/hinges) to minimise the weaker points (e.g. gaps between sections/pieces and hinges) of the armoury assembly **100**.

(24) Moreover, it is common in the prior art to provide hinges and pin-like elements to connect those two half-pipe together. However, such solution is less optimal compared to the present case where the channel **30b** having a smaller diameter is provided to receive wire or strand element **75**, wherein the wire or strand element **75** is arranged in such a way to exert a compressing force radially along the longitudinal axis of the armoury assembly **100**. FIG. **3a** shows a perspective view of a third embodiment of the armoury assembly **100** according to the present invention, wherein the armoury assembly **100** comprises at least two layers, one being an energy-absorption matrix **20** (not shown), the other **10** is located at the outermost layer of the armoury assembly **100**, wherein said layer **10** being made of a metal, an alloy or a fibre reinforced polymer, having a thickness less than the energy-absorption matrix **20**. It can be seen in this figure that a plurality of longitudinal

channels **30** are being provided to the armoury assembly **100**.

(25) FIG. **3b** is a perspective view of the third embodiment where the armoury assembly **100** is longitudinally cut into a half section. As can be seen in the FIG. **3b**, the channels **30** are substantially parallel to the longitudinal axis of the wire or strand element **75** and/or the elements to be protected (e.g. load-carrying elements).

(26) To this end, it can easily be foreseen that all or most of the channels **30** can be provided to the energy absorption matrix **20** to accommodate the wire or strand element **75**, wherein the wire or strand element **75** are arranged in such a way to exert a compressing force radially along the longitudinal axis of the armoury assembly **100**. Of course, in other embodiments, only some of the channels can be provided to house the wire and strand element **75** and the rest of the channels can be provided to house the structural material **115** or load-carrying elements **85** including strand sheathing **135**.

(27) A plan view of the third embodiment is represented in FIG. **3c**. A plurality of channels **30** are provided to the armoury assembly **100**. Some of the channels **30a** are provided to accommodate load-carrying elements **85** (shown in this embodiment are **28** channels **30a** in the central position) while the rest of the channels **30b** are provided to accommodate wire or strand elements **75** (shown in this example are nine channels **30b** in the central position and six channels **30b** in the periphery). Each of these channels **30** can further be encircled by an inner layer **40**, wherein the material for such inner layer **40** can be similar to the material for the outer layer **10**.

(28) Moreover, it is disclosed herewith that the inner layer **40** described in the FIG. **3c** can be similar to the inner layer **40** as described in the FIG. **2a**, wherein the inner layer **40** can be provided to the channels **30a**, **30b** for accommodating structural material **115** and/or load-carrying elements **85**. The thickness of the inner layer **40** may range from about 0.5 mm to 10 mm, typically from about 1 mm to 5 mm, preferably from about 2 mm to 3 mm or most preferred about 2.5 mm. To this end, it is mentioned that when the inner layer **40** is substantially a circular form, it typically has a diameter ranging from 10 mm to 50 mm, preferably between 20 mm and 30 mm.

(29) The armoury assembly **100** of this third embodiment can be used to protect the load-carrying elements **85**, as illustrated in FIG. **4**. The load-carrying elements **85** described herein can for instance be tensile elements. The load-carrying elements may have a surface area of about 150 mm² and can further be protected by a strand sheathing **135** such as HDPE, before being accommodated into the channels **30**. Of course, it can be foreseen that an inner layer **40** in form of a pipe can also be provided to the channel **30**, before accommodating the load-carrying elements **85** therein. Although only four load-carrying elements (from the front row in FIG. **4**) are shown to be protected by the strand sheathing **135**, it can be foreseen that all of them (or only some of them) can be protected by the strand sheathing **135**.

(30) To this end, it is disclosed that the armoury assembly **100** of the present invention in all embodiments may further comprise an intermediate connecting component **60** provided to the energy absorption matrix **20**. Such intermediate connecting component **60** is illustrated for example in the FIG. **2a**. The intermediate metal component **60** may be arranged to mechanically connecting an inner layer **40** and an outer layer **10** of the armoury element **100** (or connecting only to the outer layer **10**) to increase the mechanical strength of the armour assembly **100**.

(31) It is reiterated herewith that in all embodiments, the channels **30**, in particular the channel **30b** having a smaller diameter provided to the energy absorption matrix **20** for accommodating wire and strand elements **75**, can be provided either axially or helically around the armoury assembly **100** such that the wire or strand elements **75** accommodated therein can also be extended axially or helically along the armoury assembly **100**, such as to be tightened to exert a compressing or tensioning force radially towards the armoury assembly **100**.

(32) All variants of the embodiments of the armoury assembly **100** according to the present invention are capable of protecting structural material and/or load-carrying elements from various threats such as fire, TNT cutting charge (e.g. diamond charge, detonating rope and etc.), TNT blast

load for instance 0.5 meter away from elements to be protected and/or mechanical or thermal cutting threats.

(33) Specifically, the armoury assembly of the present invention have been tested and have shown it is capable of withstanding fire threat (e.g. rapid rise fire test) according to the UL 1709 standard test (e.g. fire temperature: 1100° C.; duration: 60 min), or as described in the test specifications according to Post-Tensioning Institute (PTI DC45.1-18) on recommendations for stay cable design for instance. The armoury assembly as claimed herewith is also capable of withstanding at least 15 kg and/or at most 100 kg TNT cutting charge; at least 15 kg and/or at most 100 kg TNT blast load at at least 0.5 meter away from the armoury assembly. Moreover, mechanical or thermal cutting tests have been performed and proved to be able to withstand diamond charge, linear cumulative cutting charge and detonating cord assembly (PETN) which are equivalent to approximately 15 kg or even 100 kg TNT. The armoury assembly according to the present invention is also capable of withstanding 100 kg TNT for instance.

(34) Tests have shown that the armoury assembly of the present invention is effective in protecting structural material and/or load-carrying elements. For an armoury assembly to be considered to be fully effective in protecting structural material and/or load-carrying elements from the threats described herein, following values are given: During the entire fire exposure, the temperature at the vicinity of the elements to be protected shall not exceed 300° C. As for the cutting charge test, blast test and mechanical and thermal cutting test, after being exposed the threats, the ultimate capacity of the elements to be protected (e.g. load-carrying elements) shall exceed at least 50% of its guaranteed ultimate tensile strength.

(35) By “about” or “approximately” in relation to a given numerical value, it is meant to include numerical values within 10% of the specified value.

(36) The indefinite article “a” or “an” does not exclude a plurality, thus should be treated broadly.

(37) By “one or more” or “at least one” it is meant to include the whole numbers include 1, 2, 3, 4, 5 and more up to a number which can be applied and understood by a skilled person in the art.

Claims

1. An armoury assembly for the protection of a structural material and/or load-carrying element having a longitudinal axis, wherein the armoury assembly is provided longitudinally surrounding the structural material and/or load-carrying element to be protected, the armoury assembly comprising a layer of energy-absorption matrix, and an outer layer comprising a metal, an alloy or a fibre reinforced polymer having a thickness less than the layer of energy-absorption matrix, wherein the layer of energy-absorption matrix has opposite, inner and outer surfaces defining a thickness of said layer of energy-absorption matrix therebetween, wherein longitudinal first and second channels are provided to the armoury assembly, wherein the first and second channels are substantially parallel to the longitudinal axis of the structural material and/or the load-carrying element, and wherein the first channel is embedded within the thickness of the layer of energy-absorption matrix.

2. The armoury assembly according to claim 1, wherein at least one of the first and second channels has a geometry which permits threading of a single wire or strand element thereto.

3. The armoury assembly according to claim 1, wherein the energy-absorption matrix comprises a solid filler comprising at least one of concrete, polymer-concrete, ashcrete, timbercrete, the solid filler having a compressive strength of at least about 20 MPa and/or at most about 300 MPa.

4. The armoury assembly according to claim 1, wherein at least one of the first and second channels is/are configured to accommodate one or more wire or strand elements and/or load-carrying elements thereto in such a way to exert compressing forces.

5. The armoury assembly according to claim 1, wherein a pipe element is provided to at least one of said first and second channels for receiving one or more wire or strand elements accommodated

thereto, wherein said pipe element is configured such that the one or more wire or strand elements extend axially or helically along the longitudinal axis.

6. The armoury assembly according to claim 1, wherein the outer layer comprises a plurality of patch-like elements assembled, connected and tightened to each other, such as to permit later retrofit of a critical member.

7. The armoury assembly according to claim 1, further comprising an inner layer made of a metal, an alloy, or fibre reinforced polymer.

8. The armoury assembly according to claim 5, further comprising an inner layer made of metal, an alloy, or fibre reinforced polymer, wherein the pipe element is the inner layer.

9. The armoury assembly according to claim 7, wherein the inner layer surrounds longitudinally the load-carrying element and has a same thickness as the outer layer.

10. The armoury assembly according to claim 1, further comprising an inner layer made of metal, an alloy, or a fibre reinforced polymer, wherein the layer of energy absorption matrix is sandwiched between the outer layer and the inner layer.

11. The armoury assembly according to claim 1, further comprising a plurality of the first channels all having approximately a same diameter, wherein the plurality of the first channels are provided to the armoury assembly for accommodating wire or strand elements and/or additional load-carrying elements.

12. The armoury assembly according to claim 11, wherein the plurality of the first channels are arranged approximately equidistant from each other.

13. The armoury assembly according to claim 1, further comprising an inner layer, the outer layer being made of a material having a yield strength of at most about 2000 MPa and/or at least about 200 MPa, and the inner layer is made of a material having a yield strength of at most 2000 MPa and/or at least about 200 MPa.

14. The armoury assembly according to claim 1, wherein the second channel is centrally located and defined by the inner surface of the layer of energy-absorption matrix.

15. The armoury assembly according to claim 14, further comprising a plurality of the first channels that are spaced circumferentially about the second channel.

16. The armoury assembly according to claim 14, wherein a pipe is provided in the second channel.

17. A stay cable pre-fitted or retro-fitted with the armoury assembly according to claim 1.

18. A structural material of a construction or a building, comprising a structural component selected from among a column, a rod, a beam pre-fitted or retro-fitted with the armoury assembly according to claim 1.
